

CRYPTO SYLLABUS

UNIT	DESCRIPTION	CONTACT HOURS
I	Introduction to Cryptography, OSI security architecture: Security Services, Mechanisms and attacks- Phishing, Ransomware, DoS attack. Network security model. Classical Encryption techniques - Symmetric cipher model, substitution techniques, transposition techniques. Steganography.	08
II	Conventional Symmetric Key Encryption: Block ciphers and Stream Ciphers, Block Cipher Design Principles, Modes of operation, Data Encryption Standard, Advanced Encryption Standard (AES), Multiple Encryption and Triple DES. Public key cryptography: Principles of public key cryptosystems-The RSA algorithm-Key management - Diffie Hellman Key exchange - Elliptic curve arithmetic - Elliptic curve cryptography	10
III	Hash Functions and MAC: Properties of hash functions, birthday attack, hashcash, Message Authentication Code Algorithms, MAC protocols: HMAC, CMAC. (<i>Handwritten note: & Extra SHA</i>) Digital Signatures: Classification of signature schemes: RSA signature, Digital Signature Standard, Overview of ElGamal and Schnorr schemes, One time signature schemes, Attacks on Digital Signatures, Blind Signatures.	10
IV	Introduction to Cyber Security: Email Security: Security Services for email, Attacks possible through email, Establishing keys privacy, authentication of the source, Message Integrity, Non-repudiation, Pretty Good Privacy, S/MIME. IP Security: Overview of IPSec, IPv4 and IPv6, Authentication Header, Encapsulation Security Payload (ESP), Internet Key Exchange. Transport Level Security: SSL/TLS Basic Protocol, computing the keys, client authentication, PKI as deployed by SSL, Attacks fixed in v3, Exportability, Encoding, Secure Electronic Transaction (SET).	10
V	Cyber Crimes and Cyber Security: Cyber Crime and Information Security – classifications of Cyber Crimes – Tools and Methods – Password Cracking, Key loggers, Spywares, SQL Injection – Network Access Control – Cloud Security – Web Security – Wireless Security	08